

Measure entropy

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Let (X, T, μ) be a measure preserving system.

Entropy is an important invariant

i). Two isomorphic systems has the same entropy.

used to show that Bernoulli shift $(\frac{1}{2}, \frac{1}{2})$, $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ not iso.

ii). Given (X, T) "nice situation" a unique measure of maximal entropy for (X, T) .

If a space X is equally divided into $p = 2^n$ parts. Knowing

which part the point is, gives information $n = \log_2 p$.

n digits 0, 1. Or $\log p$.

Ex. Circle rotation $(S^1, R_\alpha, \text{Leb})$. $h_\mu(R_\alpha) = 0$

Doubling map. (S^1, f_2, Leb) $h_\mu(f_2) = \log 2$

(S^1, f_2) has a lot invariant measures, Leb has maximal entropy.

Def. (Partition finite.) A finite measurable partition

$\mathcal{P} = \{P_1, \dots, P_k\}$ of X is that

$$\mu(P_i \cap P_j) = 0, \quad \mu(X - \bigcup P_i) = 0$$

Def. $\mathcal{P}, \mathcal{Q}$. $\mathcal{P} \vee \mathcal{Q} = \{P_i \cap Q_j, i, j\}$. finer partition

Def. (Entropy of a partition) For a partition

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$\mathcal{P} = \{P_1, \dots, P_k\}$ and a measure, we define

$$H_\mu(\mathcal{P}) = \sum_i -\mu(P_i) \log \mu(P_i) \quad (\text{why?})$$

Def. (Conditional entropy). Let $\mathcal{A}, \mathcal{B}$ be two partitions, then

$$\begin{aligned} H_\mu(\mathcal{B} | \mathcal{A}) &= \sum_i \mu(A_i) H_{\mu|_{A_i}}(\mathcal{B}) \\ &= \sum_i \mu(A_i) \sum_j -\frac{\mu(B_j \cap A_i)}{\mu(A_i)} \log \frac{\mu(B_j \cap A_i)}{\mu(A_i)} \end{aligned}$$

lem. Let α, β, γ be three partitions, then

- i) $H_\mu(\alpha \vee \beta) = H_\mu(\alpha) + H_\mu(\beta | \alpha)$
- ii). $H_\mu(\beta | \alpha) \leq H_\mu(\beta)$
- iii). $H_\mu(\alpha \vee \beta | \gamma) = H_\mu(\beta | \gamma) + H_\mu(\alpha | \beta \vee \gamma)$
- iv) $H_\mu(\alpha | \beta \vee \gamma) \leq H_\mu(\alpha | \beta)$

Key ingredient: Concavity of function $\phi(x) = -x \log x$.

for any $0 \leq t_i$ ~~and~~ $\sum t_i = 1$.

$$\phi(\sum t_i x_i) \geq \sum t_i \phi(x_i)$$

The function is strictly concave. equality holds only if x_i are the same.

Cor. If $\mathcal{P} = \{P_1, \dots, P_k\}$, then $H_\mu(\mathcal{P}) \leq \log k$.

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and equality holds when $\mu(P_1) = \dots = \mu(P_k) = \frac{1}{k}$.

Pf (1) definition

$$(2) \quad H_\mu(\beta|\alpha) = \sum_i \sum_j \frac{\mu(A_j)}{\mu(A_j)} - \frac{\mu(B_i \cap A_j)}{\mu(A_j)} \log \frac{\mu(B_i \cap A_j)}{\mu(A_j)}$$

$$\leq \sum_i -\mu(B_i) \log \mu(B_i)$$

$$x_j = \frac{\mu(B_i \cap A_j)}{\mu(A_j)} \quad t_j = \mu(A_j)$$

□

Definition of entropy

Notice that

$$H_\mu(\alpha \vee \beta) = H_\mu(\alpha) + H_\mu(\beta|\alpha) \leq H_\mu(\alpha) + H_\mu(\beta)$$

lem. (Subadditive sequence) If $(a_n)_n$ a positive subadditive sequence, that is $a_{m+n} \leq a_m + a_n$, then the limit exists

$$\lim_{n \rightarrow \infty} \frac{a_n}{n}$$

Introduce a notation

$$\bigvee_{j=k}^{\ell} T^j \mathcal{P} = T^{-k} \mathcal{P} \vee \dots \vee T^{-\ell} \mathcal{P}$$

lem. Let $\mathcal{P}$ be a partition, then the limit exists and called ~~ent~~

$$h_\mu(T, \mathcal{P}) = \lim_{n \rightarrow \infty} \frac{1}{n} H_\mu(\mathcal{P} \vee T^{-1} \mathcal{P} \vee \dots \vee T^{-(n-1)} \mathcal{P})$$

~~entropy of~~

Pf. let $a_n = H_\mu(P \vee T^{-1}P \vee \dots \vee T^{-(n-1)}P)$ (4)

$$= H_\mu((P \vee \dots \vee T^{-(n-1)}P) \vee (T^{-n}P \vee \dots \vee T^{-(n-1)}P))$$

$$\leq H_\mu(P \vee \dots \vee T^{-(n-1)}P) + H_\mu(\bigvee_{i=0}^{n-1} P)$$

Since μ is T -inv

$$= H_\mu(\bigvee_0^{n-1} P) + H_\mu(\bigvee_0^{n-m-1} P) = a_n + a_{n-m}.$$

We can apply the lem of subadditive sequence.

□

Def. Entropy.

$$h_\mu(T) = \sup_{\substack{P \\ \text{finite partition}}} \{ h_\mu(T, P) \}$$

Hard to compute, Sinai's thm will tell us it suffices to compute generating partition.

Example. Recall $f_2: \pi \rightarrow \pi$ let $P = \{[0, \frac{1}{2}), [\frac{1}{2}, 1)\}$
 $x \mapsto 2x$

Then $\bigvee_0^{n-1} P = \bigsqcup_{0 \leq k \leq 2^n-1} [\frac{k}{2^n}, \frac{k+1}{2^n})$

Hence.

$$\begin{aligned} h_\mu(T, P) &= \lim_{n \rightarrow \infty} \frac{1}{n} H_\mu(\bigvee_0^{n-1} P) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{0 \leq k \leq 2^n-1} -\frac{1}{2^n} \cdot \log \frac{1}{2^n} \\ &= \log 2. \end{aligned}$$

Indeed $h_\mu(T) = \log 2$. See later

Thm (Kolmogorov - Sinai) Entropy is invariant
under measurable isomorphism.

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$$\begin{array}{ccc} X_1 & \xrightarrow{T_1} & X_1 \\ \pi \downarrow & \circlearrowright & \downarrow \pi \\ X_2 & \xrightarrow{T_2} & X_2 \end{array}$$